



AHB File Reader Bus Master Technical Reference Manual

IP Solutions Division

© Copyright ARM Limited 2001. All rights reserved.

Abstract

This document provides a reference specification for the AHB File Reader Bus Master. The AHB File Reader Bus Master is a component for the Micropack EASY system. The function of this component is to generate a set of AHB bus transactions from a stimulus file. The benefits of the AHB File Reader Bus Master to an AMBA system designer are that it facilitates fast system simulations, it is simple to demonstrate multiple master systems and it provides a reference design of a fully functional AHB master.

Keywords

AMBA, AHB, Bus Master, File Reader, EASY, Micropack

Contents

1	ABOUT THIS DOCUMENT	3
1.1	References	3
1.2	Terms and abbreviations	3
2	INTRODUCTION	4
3	FUNCTIONAL OVERVIEW	5
3.1	AHB File Reader Bus Master	5
3.1.1	Pin List	5
3.1.2	AHB Interface	6
3.1.3	AHB File Reader Bus Master Function	6
3.2	AHB-Lite File Reader	6
3.2.1	Pin List	6
3.2.2	AHB-Lite Interface	7
3.2.3	Input Stimulus File	8
3.2.4	File pre-processing	10
3.2.5	AHB-Lite File Reader Function	11
3.3	AHB Lite to AHB Wrapper	13
3.3.1	Pin List	13
3.3.2	AHB Interface	13
3.3.3	AHB-Lite Interface	13
3.3.4	AHB Lite to AHB Wrapper Function	13
4	SYNTHESIS AND TEST	14
4.1	Synthesis	14
4.2	Test	14

1 ABOUT THIS DOCUMENT

1.1 References

Ref	Doc No	Author(s)	Title
[1]	SC056 TRM 0000 A01	A Harris	AHB Example Bus Master TRM
[2]	ARM IHI 0011A	ARM	AMBA Specification (Rev 2.0)
[3]	SC061 SPEC 0001 A01	B Mathewson	AMBA Compliance Testbench Structure
[4]	SCB00 DESR 0000 D	N Bray	ARM HDL Coding standard for Reusable IP
[5]	AMBA DISC 0007 A-03	B Mathewson	AHB-Lite Specification

1.2 Terms and abbreviations

Term	Meaning
AMBA	Advanced Microcontroller Bus Architecture
AHB	Advanced High-performance Bus
AHB-Lite	Advanced High-performance Bus Lite (Single Master AHB)

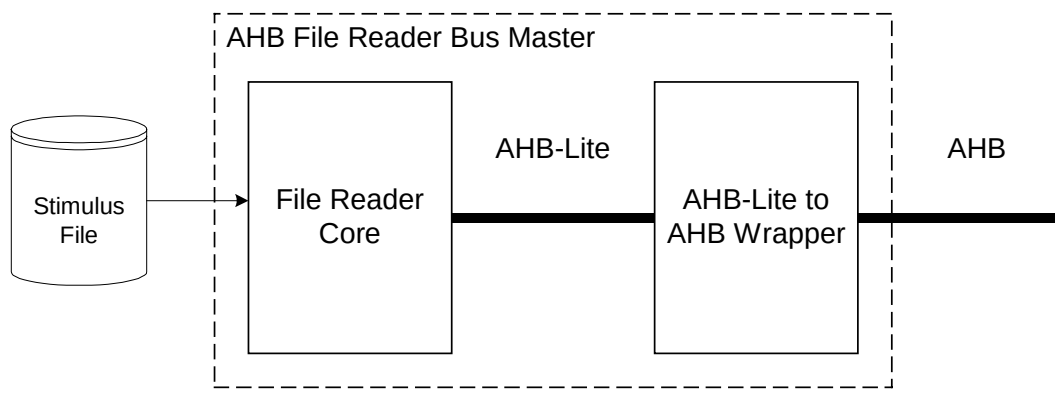
2 INTRODUCTION

The AHB File Reader Bus Master is intended to allow designers to simulate AMBA systems quickly and efficiently by using it to generate explicit bus transfers. The master will operate in the existing Micropack EASY system with or without an ARM core present. The File Reader can be used as a Bus Functional Model (BFM) facilitating arbiter algorithm design by being able to easily create multiple master systems where the designer has direct control over the bus activity. The AHB File Reader Bus Master is split into two parts: the AHB-lite File Reader and an AHB-Lite to AHB Wrapper. The AHB-Lite interface is a sub-set of the AHB interface. Differences between AHB and AHB-Lite interfaces are:

- no arbitration - the AHB-Lite master always has control of the bus.
- no SPLIT or RETRY slave responses, the AHB-Lite master only has to deal with a slave ERROR response.

Using the AHB-Lite interface makes the bus transfers generated by the AHB-Lite File Reader easier to understand, hence easier to debug.

The AHB Wrapper is used to expand the AHB-Lite to a fully compliant AHB interface. This is the same wrapper used by another EASY component, the AHB Example Bus Master. For a functional description and list of signals for the AHB-Lite to AHB Wrapper, the reader is referred to the AHB Example Bus Master Technical Reference Manual[1].



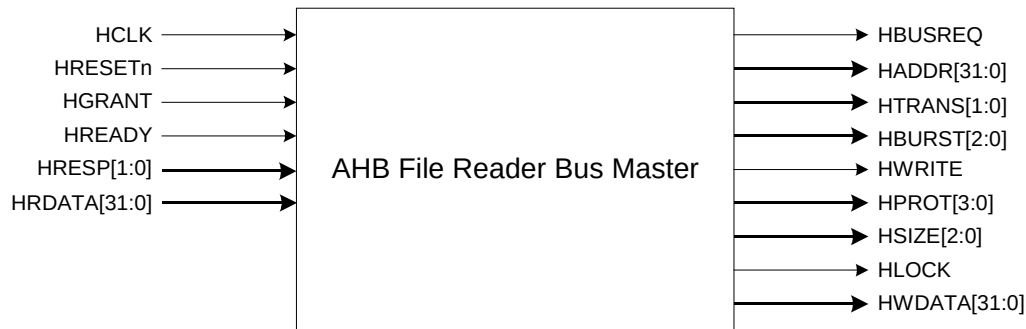
The AHB-Lite File Reader performs all AHB burst types at data widths of 8, 16 and 32 bits, inserts BUSY states during bursts and performs idle transfers. The data received during read operations is not stored in the AHB-Lite File Reader, but it does compare data received with the expected data and reports differences during simulation.

The AHB-Lite File Reader is controlled entirely via the stimulus file at simulation run time. It does not have a slave interface, therefore cannot be addressed by another AHB master.

3 FUNCTIONAL OVERVIEW

3.1 AHB File Reader Bus Master

The AHB File Reader Bus Master connects the AHB-Lite File Reader and the AHB-Lite to AHB Wrapper together via an AHB-Lite bus interface. The external interface is then AHB only.



3.1.1 Pin List

Inputs

Pin Name	Width	Description
HCLK	1	AHB clock. All signal timings are related to the rising edge of HCLK.
HRESETn	1	AHB active low reset, used to reset system and the bus.
HGRANT	1	Grant signal indicates the File Reader Bus Master is currently the highest priority. Bus master changes only occur at the end of the current transfer, i.e. both HREADY and HGRANT asserted.
HRDATA	32	Transfers data from the addressed slave to the File Reader during read operations.
HREADY	1	When asserted by the addressed slave indicates the current transfer has completed, when de-asserted the slave forces the transfer to be extended.
HRESP	2	Enables the slave to indicate the transfer status, and relinquish the bus if required.

Outputs

Pin Name	Width	Description
HBUSREQ	1	When asserted indicates the File Reader is requesting use of the bus.
HLOCK	1	When asserted indicates the File Reader requires uninterrupted access to the bus.
HADDR	32	32-bit system address bus.
HTRANS	2	Indicates the transfer type: a new burst, part of a burst, idle or busy
HWRITE	1	When asserted indicates a write transfer, when de-asserted indicates a read transfer.
HSIZE	3	Indicates the width of the current data transfer, when used with HADDR indicates which byte lanes are active.
HBURST	3	Indicates when the File Reader is performing a transfer that is part of a burst. 4, 8 and 16 beat fixed length bursts are supported.
HPROT	4	Protection control signals for the current transfer.
HWDATA	32	Transfers data from the File Reader to the addressed slave during write operations

3.1.2 AHB Interface

The AHB interface for an AHB bus master is described in the AMBA Specification[2].

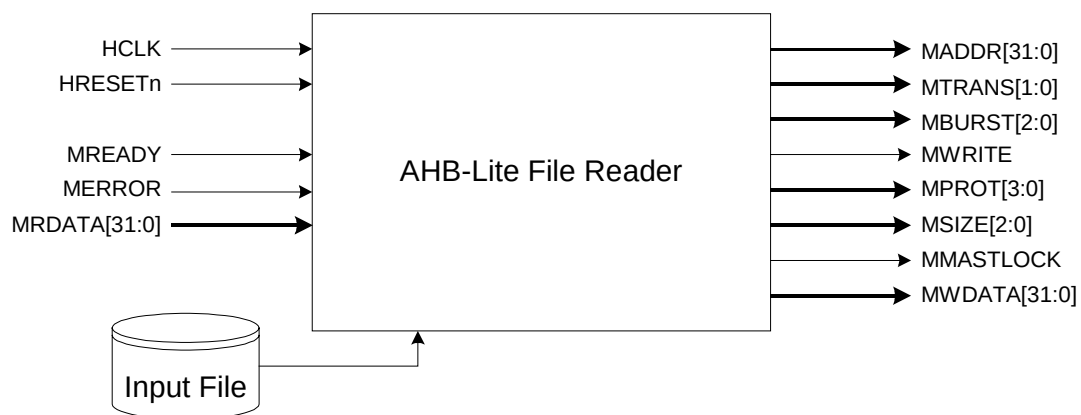
3.1.3 AHB File Reader Bus Master Function

The AHB File Reader is a structural block. Its function is to:

- connect the AHB-Lite File Reader and AHB Wrapper together via an AHB-Lite interface.
- connect the AHB Wrapper to the External AHB interface.

3.2 AHB-Lite File Reader

The File Reader has the function of interpreting the input file and generating a sequence of transfers on the AHB-Lite interface. The File Reader does not have any inputs or outputs other than the AHB-Lite interface.



3.2.1 Pin List

Inputs

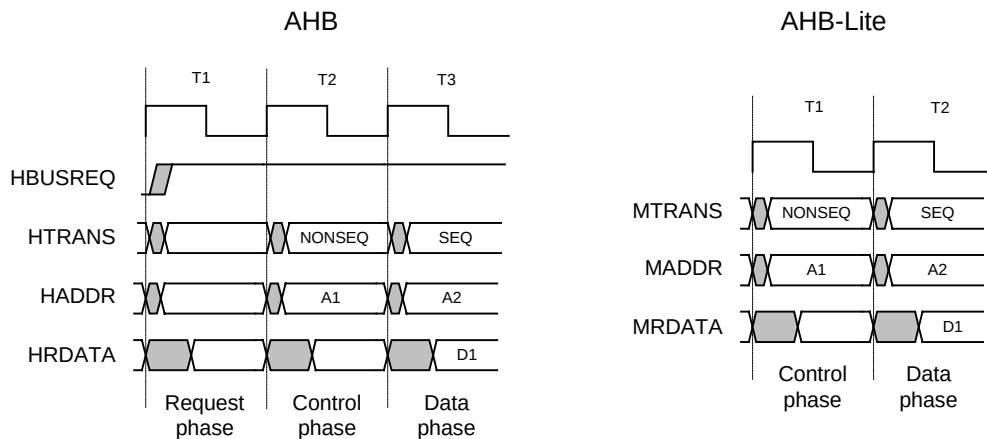
Pin Name	Width	Description
HCLK	1	AHB clock. AHB-Lite interface and File Reader are synchronous with the AHB system
HRESETn	1	AHB active low reset. AHB-Lite interface and File Reader are reset by the AHB reset signal.
MRDATA	32	AHB Read Data passed through the wrapper to the AHB-Lite interface, current data corresponds to the previous address on MADDR.
MREADY	1	When asserted by the AHB Wrapper indicates the current transfer has completed, when de-asserted the Wrapper forces the transfer to be extended..
MERROR	1	When asserted indicates an Error response from the addressed AHB slave.

Outputs

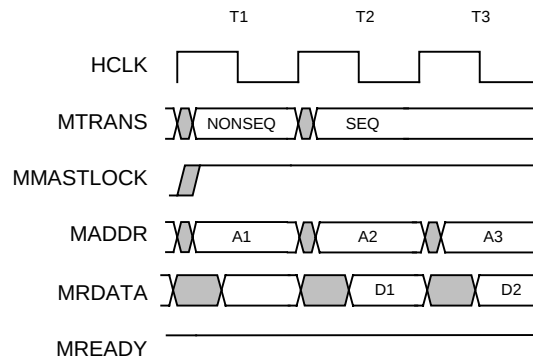
Pin Name	Width	Description
MMASTLOCK	1	When high indicates that a locked transfer is required, this signal must be asserted a cycle before MTRANS changes from idle to non-sequential.
MADDR	32	Indicates the required address for the transfer. MADDR is subject to the same restrictions as HADDR [2]. i.e. Address alignment.
MTRANS	2	Transfer type, indicates the state of the current transfer. MTRANS assumes the same states as AHB signal HTRANS [2].
MWRITE	1	Data direction, indicates a write operation when high and a read operation when low. MWRITE has the same function as HWRITE [2].
MSIZE	3	Transfer size, indicates the active data lanes for the current transfer. MSIZE assumes the same states as AHB signal HSIZE [2].
MBURST	3	Burst type, indicates the type of burst for the current transfer. MBURST assumes the states as AHB signal HBURST [2].
MPROT	4	Protection control signals for the current transfer. MPROT assumes the states as AHB signal HPROT [2].
MWDATA	32	Data from the file reader, the current data corresponds to the previous address on MADDR.

3.2.2 AHB-Lite Interface

The AHB-Lite interface[5] does not have a Request phase since in a single master system the AHB-Lite master always has control of the bus.



The start of a transfer is signalled by MTRANS becoming NONSEQ. The AHB-Lite to AHB wrapper is responsible for generating the required AHB request signal. For locked transfers in an AHB system HLOCK is asserted during the Request phase. The AHB-Lite File Reader asserts MMASTLOCK in its Control phase, so it has the same timing as the MTRANS output.



The AHB-Lite to AHB wrapper is responsible for generating the required AHB lock signal. All other AHB-Lite signals operate as described in the AHB Example Bus Master TRM[1].

3.2.3 Input Stimulus File

The filename of the stimulus data file is specified using a generic / parameter at the point of instantiation in the HDL code.

The syntax of the input stimulus file has been derived from the syntax specified in the AHB Slave Compliance Testbench Structure[3] document. Some changes have been made to this syntax:

1. The addition of the `lock` field to indicate a locked transfer.
2. Removed memory test command, since this involves random behaviour.

The syntax uses a single letter for each command followed by a number of fields.

Command Syntax

Cmd	Fields						
W	Address	Data	[Size]	[Burst]	[Prot]	[Lock]	
R	Address	Data	[Mask]	[Size]	[Burst]	[Prot]	[Lock]
S	Data	[Mask]					
B							
I	[Address]	[Dir]	[Size]	[Burst]	[Prot]	[Lock]	
P	Address	Data	[Mask]	[Size]	[incr sing]		[Prot]
L	Number						

W - The write command starts a write burst and may be followed by one or more S vectors. The number of S vectors is set by the `size` and `burst` fields for fixed length bursts. There is no limit to the number of S vectors for undefined length bursts, as long as it does not cause the address to cross a 1k boundary.

R - The read command starts a read burst and may be followed by one or more S vectors. The number of S vectors is set by the `size` and `burst` fields for fixed length bursts. There is no limit to the number of S vectors for undefined length bursts, as long as it does not cause the address to cross a 1k boundary.

S - The sequential vector provides data for a single beat in the burst. The File Reader calculates the address required.

B - The busy command inserts a BUSY transfer mid burst. An INCR burst is allowed to have a busy after its last transfer (while the master determines whether or not it has another transfer to complete). It is not valid to have a busy command when a burst is not in progress.

I - The Idle command performs an idle transfer. The options allow you to set up the control information during the idle transfer, and to specify if the transfer is locked or unlocked.

P - Poll command performs a read transfer that repeats until the data matches the required value. The poll vector can only be used for INCR or SINGLE burst types.

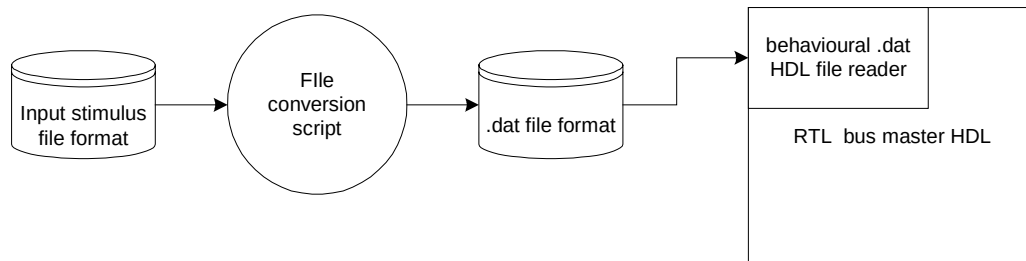
L - Loop command repeats the last command a number of times. An L command can only follow a W or R when the burst type is INCR or SINGLE.

Command Fields

Field	Default	Values	Description
Address	0x00000000	A 32-bit hex value	First address of burst. Compulsory field for read, write and poll commands.
Data	N/A	A 32-bit hex value	Required data value. Compulsory field for read, write, sequential and poll commands.
Mask	0xFFFFFFFF	A 32-bit hex value	Bit mask, allows read data to be masked when tested against the required data.
Size	Size32	b byte h hword w word size8 size16 size32	Data size used for read or write transfers.
Burst	Incr	sing single incr incr4 wrap4 incr8 wrap8 incr16 wrap16	Burst type used for read or write transfers.
Prot	0000	A 4-bit binary value	Direct representation of the MPROT value for the transfer.
Lock	Nolock	nolock lock	Indicate if the transfer is locked or not
Dir	Read	read write	Indicates the value of MWRITE during an Idle command.
Number	N/A	1 to 1023	Loop repeat value. Max value defined by number of bits used for of counter in File Reader.
Comment Delimiter		; # // --	all common comment delimiters are valid.

3.2.4 File pre-processing

The stimulus file format is converted using a script into a format that can be fed directly into the HDL code. The script is used to ensure the syntax of the input file is correct, and offers useful error messages when the syntax is not correct.



Additional checks performed by the script to ensure correct File Reader operation:

Error-checks (file conversion aborted):

- Input file is readable (exists and has correct permissions)
- No unrecognised commands within the file
- No required fields that are missing or in the wrong format
- No S vectors before a defined-length or undefined-length burst has started
- No B vectors before a defined-length or undefined-length burst has started
- For Poll commands, burst types are either incr or single.
- No incompatible commands inside a burst (Write, Read, Idle or Poll)
- Loop command has number specified
- Loop Number field out of range [1 to 1023], set by 10 bit counter in AHB File Reader Bus Master.
- Loop number does not exceed number of remaining transfers, for each defined-length burst type.
- Size value does not exceed the data bus width. Max value is word | size32 for the AHB File Reader Bus Master.
- Burst will not exceed 1kB address boundary, for both defined and undefined-length bursts.
- Address is word-aligned, for word transfers
- Address is halfword-aligned, for halfword transfers

Warning generated if (file conversion continues):

- An optional field has an invalid value
- Number of S vectors following a W | R command is correct for a fixed length burst (a burst is terminated early). Note that this allows the simulation of early-terminated bursts.

This script does not implement any unfolding of loops or address calculations, this is part of the File Reader functionality. The conversion from text to a list of 32-bit hex numbers is a one to one mapping. The script is written in a platform independent language (Perl) and converted to a binary for Solaris and HP-UX platforms.

3.2.5 AHB-Lite File Reader Function

Write Command

- Cycle 1: The File Reader sets up the control signals from the command and MWRITE is asserted. MTRANS is set to Non-sequential indicating the start of a new transfer, the data field is stored ready to be output during the Data phase.
- Cycle 2: If MREADY is asserted then the file reader proceeds to the second Control phase. If Burst is not a single transfer then the File Reader calculates the next address based on Size and Burst values. The new address is output and MTRANS is set to sequential. This is the first Data phase where the Data is output for the previous address value.

Read Command

- Cycle 1: The File Reader sets up the control signals from the command and MWRITE is de-asserted. MTRANS is set to Non-sequential indicating the start of a new transfer.
- Cycle 2: If MREADY is asserted then the File Reader proceeds to the next Control phase. If Burst is not a single transfer then the File Reader calculates the next address based on Size and Burst values. The new address is output and MTRANS is set to sequential. The data is read for the previous address value and is compared with the Data field after applying the Mask. Differences between the expected and read data are reported to the simulation environment.

Sequential Command

A sequential command is valid once a burst transfer has been started by a Read or Write command.

- Cycle n: Address set to calculated value and MTRANS will be set to Sequential.
- Cycle n+1: For a write burst the Data field is read and output by the File Reader. The Mask field is ignored. For a read burst the Data field is compared with the input data after the Mask has been applied. Differences between the expected and read data are reported to the simulation environment.

Busy Command

A busy command is valid once a burst transfer has been started by a Read or Write command.

- Cycle n: Address set to calculated value and MTRANS will be set to Busy.
- Cycle n+1: Data is ignored.

Idle Command

The File Reader sets MTRANS to Idle and sets the control signals to the default values except for those with values specified in the command.

Poll Command

The File Reader sets up a read of a single Address and reads in the data, on the following cycle. If the data read matches the Data field then the File Reader will continue to the next command, otherwise the File Reader will repeat the read from the same address.

Loop Command

The File Reader is responsible for storing the last command and being able to repeat it a number of times. The File Reader constrains the field value to 1024 because it uses a 10-bit counter.

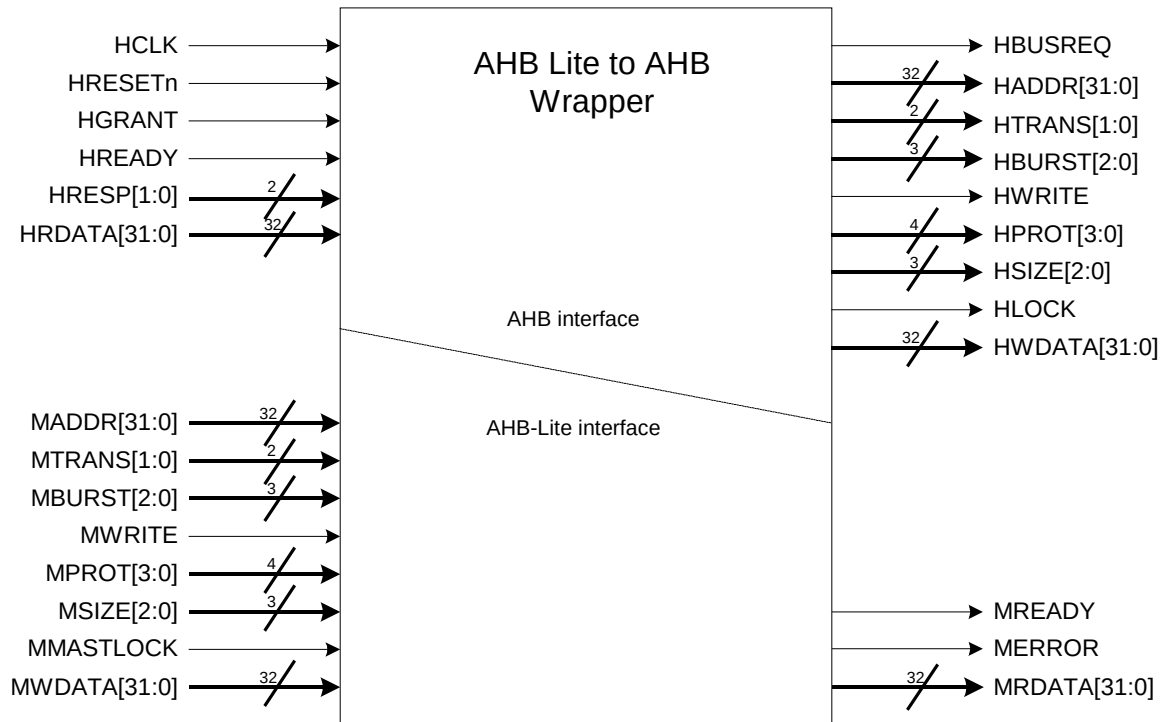
The File Reader is synchronous with the AHB bus clock signal, HCLK and is reset by the AHB reset signal HRESETn.

In accordance with the current Micropack EASY system the File Reader will be little-endian. However comments in the HDL code will explicitly show how to make the File Reader big-endian if it should be required. For documentation on endianness the reader is referred to the AMBA specification[2] Tables 3-6 and 3-7.

When MERROR is asserted indicating an error response from the Slave, the File Reader continues with the current transfer. No attempt is made to re transmit the erroneous transfer. The Slave error response is reported to the simulation environment.

3.3 AHB Lite to AHB Wrapper

The AHB Wrapper converts an AHB-Lite interface into a full AHB interface.



3.3.1 Pin List

The AHB Lite to AHB Wrapper pin list is described in AHB Example Bus Master Technical Reference Manual[1].

3.3.2 AHB Interface

The AHB interface for an AHB bus master is described in the AMBA Specification[2].

3.3.3 AHB-Lite Interface

The AHB-Lite interface is described in AHB Example Bus Master Technical Reference Manual[1].

3.3.4 AHB Lite to AHB Wrapper Function

The Wrapper function is described in AHB Example Bus Master Technical Reference Manual[1].

4 SYNTHESIS AND TEST

4.1 Synthesis

The AHB File Reader Bus Master is designed so that where ever possible RTL code is used to describe its logic[4]. All RTL code is written for synthesis. The AHB-Lite File Reader requires some behavioural code for input file reading. This code will be enclosed in pragmas to allow the block to pass through synthesis tools.

The synthesizable logic should conform to the AMBA timing parameters:

Outputs valid 40% of cycle after rising edge of HCLK.

Inputs valid 30% of cycle before rising edge of HCLK.

4.2 Test

The AHB File Reader Bus Master is only intended for simulation. Manufacture tests are not a requirement.